



Automated Mild Cognitive Impairment Detection (PD-MCIx) in patients with Parkinson's Disease

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Background

- **Parkinson's Disease (PD):** A progressive neurodegenerative disorder caused by dopamine neuron loss in the substantia nigra.^{1,2}
- **Mild Cognitive Impairment (MCI):** Marked by mild cognitive deficits, considered a transitional phase between normal aging and dementia.³
 - Subtypes: **amnesic** or **non-amnesic**.⁴
- Diagnosing MCI in PD (PD-MCI) for research typically requires clinical expertise from licensed professionals.
- **Current** auto-diagnostic tools:
 - **machine learning** primarily used for cognitive assessment analysis⁵, not MCI diagnosis
 - **natural language processing** to detect language impairments linked to cognitive decline⁶
 - **signal processing** to extract key features from voice recordings
 - require **prior technical expertise** to operate effectively⁶ and rigid data input requirements limited their usability and application
- New modifiable tools needed to streamline and increase accuracy for MCI diagnosis for research

Goal: To develop a beginner-friendly, customizable script that diagnoses **mild cognitive impairments** for research following the Movement Disorder Society Guidelines for **PD-MCI diagnostic criteria**⁷ using cognitive, subjective, and functional data

Methods

Parent Study#1 (N=99 PD):

Relationship between daily exercise with brain activity/ cognitive performance in Parkinson's Disease.

Demographics and Characteristics of Study Samples

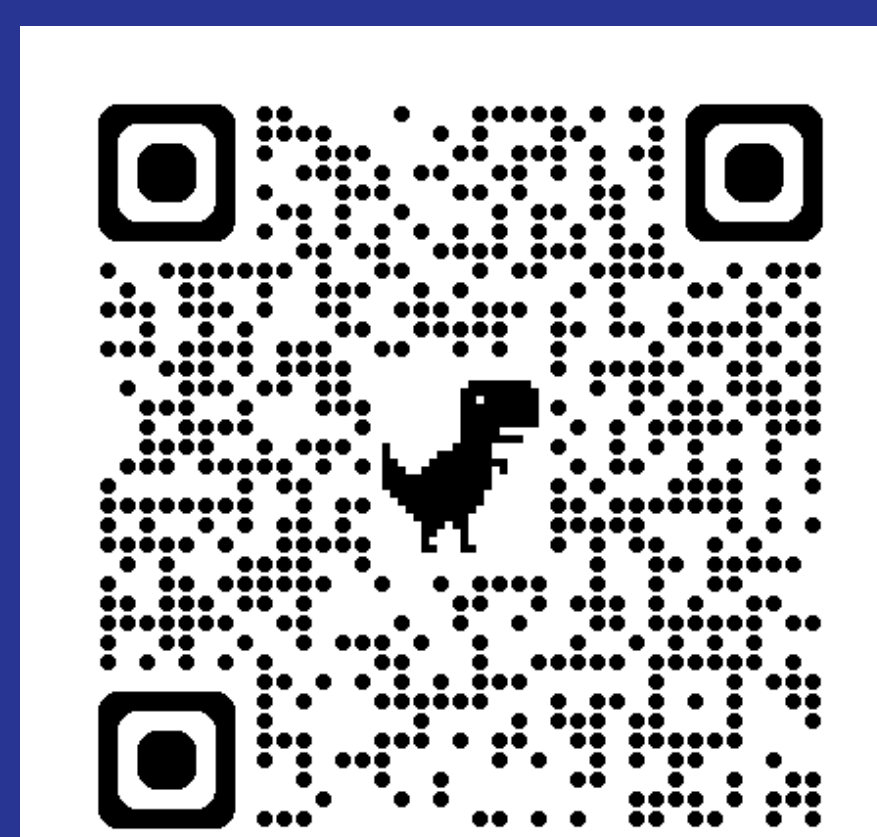
Category	Variable	Parent Study #1	Parent Study #2
General	Age (years)	66.7 (8.8)	67.7 (9.3)
	Education (years)	15.5 (1.2)	16.2 (3.2)
	Sex (#M/F)	57/42	119/52
	Disease Duration (mo)	57 (58.1)	63.3 (58.9)
	LED (mg/day)	478.1 (366.1)	714.6 (682.7)
Ethnicity	Hispanic	7	5
	Non-Hispanic	90	166
Race	AI/A/B/NH/W *	3/2/1/7/0/86	2/3/1/3/1/160

*AI =American Indian/ A = Asian /B = Black/ NH =Native Hawaiian or Pacific Islander/ W = White

Parent Study#2(N= 181 PD):

Novel Predictors of Cognitive and Psychiatric Change in Parkinson's Disease.

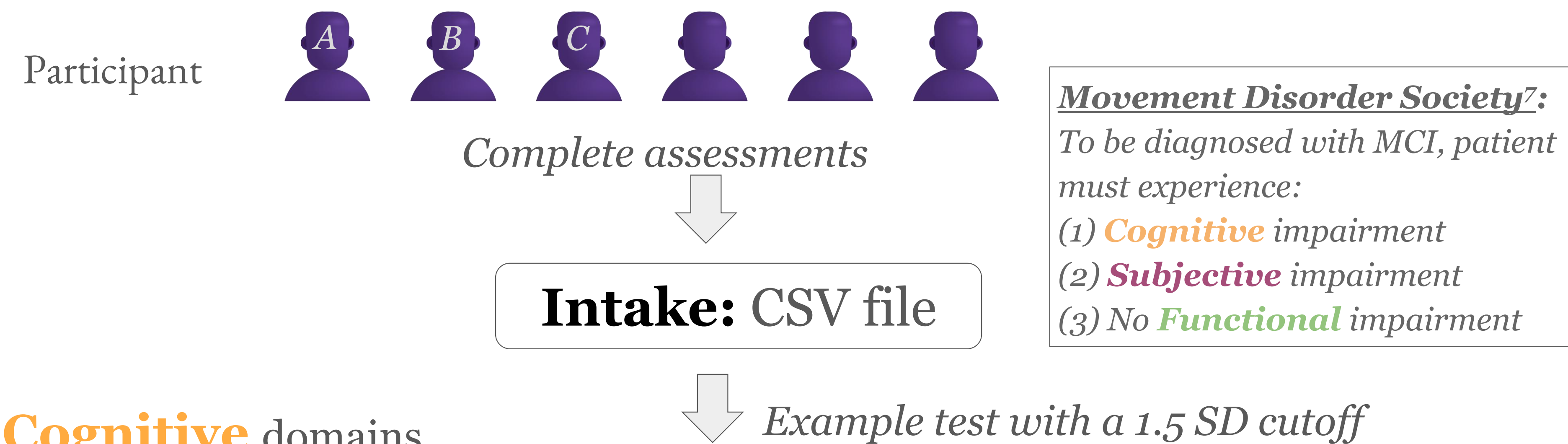
WEBSITE



PD-MCIx Interface

Accessible in Python, R, and on website

Autodiagnoser for PD-MCI (PD-MCIx)



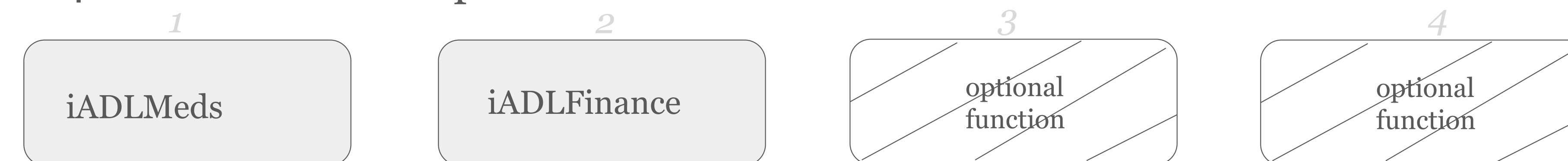
5 Cognitive domains

	Memory	Attention	Executive Func.	Visuospatial	Language
Test 1	CVLT Free Recall SS	DOT-A Span t-score	DKEFS Switch Tot z-score	Judgement of Line t-score	DKEFS Cat. Fluency t-score
Test 2	Brief Memory Test z-score	CWIT Color Name SS	WCST Per. Er.. z-score	Hooper Visual SS	Boston Naming z-score

1 to 4 Subjective Reports



1 to 4 Functional Reports



Output: CSV file

Participant	Cognitively impaired	Subjectively Impaired	Functionally Impaired	PD-MCIx	subtype
A	0	1	0	noMCI	n/a
B	1	1	0	MCI	Amnesic
C	1	1	1	noMCI	n/a

Summary and Conclusions

- 1) **Streamlines** the diagnostic process for research & highly adaptable
- 2) **Open source code** and publicly available **website**
- 3) Comparable to clinician-reviewed diagnosis, but conservative **under diagnosis** (static rule-based logic)

- ❑ **Two part validation** was done using Cohen's Kappa and a proportion test
- ❑ **Substantial agreement between PD-MCIx and the clinical diagnosis**, indicating that the automated classification aligned with clinician reviewed diagnoses

Future Directions

- Future work could incorporate **additional data types** (e.g., neuroimaging, speech samples, to enhance diagnostic accuracy and subtype differentiation.
- Validate the tool against **clinical diagnosis** + **integrate clinician feedback** mechanisms directly into the tool to continuously refine diagnostic criteria and feature selection
- Potential to be used for other populations (Alzheimers)

Sources

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